



COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Assignment 2 – FALL 2025

Course Title:	Digital System Design	Course Code:	CPE 344	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais	Program Name:	BS Computer Engineering		
Semester:	7 th	Batch:	FA22	Section:	A, B
Date Given:	10 th October, 2025			Submission Date:	13 th October, 2025
Maximum Marks:	20			Name:	
Registration Number:					

Important Instructions / Guidelines:

- Do your own work, PLAGARISM will be graded as ZERO
- No late submission.

Question 1:

[CLO1-PLO3-C5]

[10 Marks]

Design a multi-function digital circuit that takes two 8-bit inputs from registers **A** and **B** and performs the following operations based on the specified conditions:

- If **A = B**: The sum of both numbers is incremented by 1 and the result is stored in **register A**.
- If **A < B**: The difference of the two numbers is decremented by 1, and the result is stored in **register B**.
- If **A > B**: Compute $4 \times A$ and store the result in **register A** and compute $B \div 8$ and store the result in **register B**.

Note: Each operation must be completed within a single clock cycle between two loads into the registers **A** and **B**. Initially, the data is loaded into registers **A** and **B** from an external source. The design should be efficient utilizing the same resources to perform multiple functions wherever possible.